Vardhan Palod

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Research Objective: My current research focuses on understanding and enhancing the reasoning abilities of Large Language Models (LLMs). In particular, my work explores: (i) the semantic properties and underlying causes of performance improvements arising from the use of intermediate tokens in language models, and (ii) the effects of reinforcement learning on task performance and the interpretability of intermediate reasoning processes. Additionally, I am actively developing robust and scalable frameworks to enable effective and trustworthy Human–AI collaboration.

Education

Ira A. Fulton School of Engineering, Arizona State University *Aug 2025 – May 2029 (Expected)*
PhD in Computer Science, Advisor: *Prof. Subbarao Kambhampati* GPA: 4.0/4.0

Ira A. Fulton School of Engineering, Arizona State University *Aug 2024 – May 2025*
Master of Science in Computer Science - converted to PhD GPA: 4.0/4.0

Birla Institute of Technology and Science, Goa, India *Aug 2020 – May 2024*
B.E. in Computer Science with a minor in Data Science CGPA: 8.56/10

Research Experience

Research Scientist Intern *May 2026 - Aug 2026*
Meta *Menlo Park, CA, USA*

- **Formulated a Meta-Evaluation Framework** combining distributional coverage, behavioral faithfulness (fidelity & adaptability), production outcome correlation, and reproducibility to **quantify user simulator reliability**.
- **Engineered persona-based and reward-aligned LLM user simulators** to model multi-turn conversational trajectories and decision-making dynamics, systematically resolving failure modes in legacy baseline models.
- **Developed an autonomous agent optimization loop via *Meta-Harness***, leveraging high-fidelity synthetic user simulators to iteratively refine agent reasoning policies, achieving a 5% to 10% improvement over baselines.

Graduate Research Associate *Jan 2025 - Present*
School of CS & AI, Arizona State University *Tempe, AZ, USA*

- Investigating language and reasoning models' reasoning abilities via fine-tuning and RL training, and developing efficient inference methods that can reduce the deployment cost of LLMs.

Publications

- **Masked Self-Distillation: Internalizing the Chain-of-Thought in Language Models**
  Durgesh Kalwar*, **Vardhan Palod***, Jaya Adithya Pavuluri, Subbarao Kambhampati. (*Equal contribution.)
  ICML 2026, Workshop on Foundations of Deep Generative Models
- **Evaluating the False Trust Engendered by LLM Explanations**
  Vardhan Palod*, Upasana Biswas*, Subbarao Kambhampati. (*Equal contribution.)
  NeurIPS 2026, Workshop on Trustworthy AI for Good
- **Beyond semantics: The unreasonable effectiveness of reasonless intermediate tokens**
  Karthik Valmeekam*, **Vardhan Palod***, Kaya Stechly*, Atharva Gundawar, Subbarao Kambhampati. (*Equal contribution.)
  Transactions of Machine Learning Research (TMLR), 2026 (J2C Certification - presentation at NeurIPS 2026)
  NeurIPS 2025, Foundations of Reasoning in Language Models Workshop
  NeurIPS 2025, Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning (LAW)
- **Performative Thinking? The Brittle Correlation Between CoT Length and Problem Complexity**
  Vardhan Palod, Karthik Valmeekam, Kaya Stechly, Subbarao Kambhampati.
  NeurIPS 2025, Workshop on Efficient Reasoning
- **Who is Helping Whom? Analyzing Inter-dependencies to Evaluate Cooperation in Human-AI Teaming**
  Upasana Biswas, **Vardhan Palod**, Siddhant Bhambri, Subbarao Kambhampati.
  AAAI 2026

- **Stop Anthropomorphizing Intermediate Tokens as Reasoning/Thinking Traces!**
  Subbarao Kambhampati, Kaya Stechly, Karthik Valmeekam, Lucas Saldyt, Siddhant Bhambri, **Vardhan Palod**, Atharva Gundawar, Soumya Rani Samineni, Durgesh Kalwar, Upasana Biswas.
  ICML 2026
  NeurIPS 2025, Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning (LAW)
  NeurIPS 2025, Workshop on Interpreting Cognition in Deep Learning Models (CogInterp)
- **Discounting and Drug Seeking in Biological Hierarchical Reinforcement Learning**
  Vardhan Palod, Pranav Mahajan, Veeky Baths, Boris S. Gutkin.
  CCN 2025 (Computational Cognitive Neuroscience Conference)

Notable Research Projects

Inference time scaling of LLMs with Rollouts Jan 2025 - Present

- Built an inference-time scaling technique for large language models (LLMs) enabling strategic lookahead through rollouts before committing to the next action in reasoning tasks. Evaluated GPT-4o-mini on the Game of 24 domain, achieving 70% accuracy, a 3x improvement over the prior 21% baseline.

Leveraging LLMs as World-models for training RL Agents Aug 2024 - Dec 2024

- Developed a framework using Llama 3.1 to generate state-action trajectories for Gridworld environments, enabling accelerated convergence of model-free RL agents to optimal policies by up to 50% and minimized unsafe exploration by up to 75% as compared to agents that are directly trained in real environments

Technical Strengths

Programming Languages: Python, C/C++, JAVA, PDDL

Frameworks: verl (GRPO and FSDP SFT), Pytorch, HuggingFace, Docker, Kubernetes, Scikit-Learn, Numpy, Pandas

Teaching & Service

Teaching:

- **Teaching Assistant - CSE 355** Jan 2025 - May 2025
  Introduction to Theoretical Computer Science by Prof. Heewook Lee

Reviewing:

- **(AAAI 27)** *The Forty-First Annual AAAI Conference on Artificial Intelligence*
- **(ICLR 27)** *The Fifteenth International Conference on Learning Representations*
- **(NeurIPS 26 AI4Good Workshop)** *Workshop on Trustworthy AI for Good*
- **(AAAI 26)** *The fortieth Annual AAAI Conference on Artificial Intelligence*
- **(NeurIPS 25 LAW workshop)** *Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning*